



Gakkou No Shiken (■■■■■)

Official REST API Specification & Mobile Integration Manual

Version 2.0 • 2026 Prometric CBT Engine Edition • Bangladesh's #1 Japanese Exam Platform

Base URL (Production API):	https://gakkounoshiken.site/api/ (Do NOT use 'www')
Base URL (Local Dev):	http://127.0.0.1:8000/api/
Authentication Scheme:	JSON Web Token (JWT) — Authorization: Bearer <access_token>
Data Format / Content-Type:	<code>application/json; charset=utf-8</code>
Target Clients:	Flutter (iOS/Android), React Native, Kotlin, Swift, Web SPA

Overview

This specification documents the complete backend API for **Gakkou No Shiken**. The API powers authentic Computer-Based Testing (CBT) simulations for **JFT-Basic (A2)** and **Specified Skilled Worker (SSW 1 & 2)** examinations. It includes real-time scoring, CEFR scaling (10–250 scaled IRT score), 10 native language translation aids, OTP email verification, Google OAuth, candidate dashboard analytics, and live national leaderboards.

1. Authentication & Candidate Accounts

Candidate authentication utilizes JWT tokens. Pass access tokens in the Authorization: Bearer <token> header. Registration is secured with a 6-digit email OTP verification step.

POST

/api/auth/send-otp/

Auth: Public (None)

Description: Validates candidate details, generates a 6-digit verification OTP code, and emails it to the user. (Expires in 15 mins).

Parameter	Type	Required	Description
username	string	Yes	Unique username for candidate.
email	string	Yes	Valid email address for OTP delivery.
password	string	Yes	Password (min 6 characters).
password_confirm	string	Yes	Must match password exactly.
first_name	string	No	Candidate given name.
last_name	string	No	Candidate surname.

Request Payload (JSON):

```
{ "username": "tanvir_japan", "email": "tanvir@example.com", "password": "Password123!", "password_confirm": "Password123!", "first_name": "Tanvir", "last_name": "Ahmed" }
```

Response Example (200/201 OK):

```
{ "status": "success", "email": "tanvir@example.com", "message": "A 6-digit verification code has been sent to tanvir@example.com." }
```

POST

/api/auth/verify-otp/

Auth: Public (None)

Description: Verifies the 6-digit OTP, creates the candidate account in the database, and returns JWT access/refresh tokens.

Parameter	Type	Required	Description
email	string	Yes	Candidate registered email.
otp_code	string	Yes	6-digit code received via email.

Request Payload (JSON):

```
{ "email": "tanvir@example.com", "otp_code": "839201" }
```

Response Example (200/201 OK):

```
{ "user": { "id": 42, "username": "tanvir_japan", "first_name": "Tanvir", "last_name": "Ahmed", "full_name": "Tanvir Ahmed", "email": "tanvir@example.com", "is_staff": false, "profile": { "bio": "", "target_exam": "jft_basic", "target_exam_display": "JFT-Basic (A2 Standard)", "japanese_level": "n4", "japanese_level_display": "Elementary (N4 / A2)", "location": "Dhaka, Bangladesh" } }, "tokens": { "refresh": "eyJhbGciOi...", "access": "eyJhbGciOi..." }, "message": "Email verified successfully! Welcome to Gakkou No Shiken, tanvir_japan!" }
```

POST /api/auth/resend-otp/

Auth: Public (None)

Description: Resends a fresh 6-digit OTP code to the candidate's email if the previous code expired or was not received.

Parameter	Type	Required	Description
email	string	Yes	Candidate registered email.

Request Payload (JSON):

```
{ "email": "tanvir@example.com" }
```

Response Example (200/201 OK):

```
{ "status": "success", "email": "tanvir@example.com", "message": "A new 6-digit verification code has been sent to your email." }
```

POST /api/auth/login/

Auth: Public (None)

Description: Authenticates a candidate using either their Username OR Email address + Password. Returns JWT tokens.

Parameter	Type	Required	Description
username	string	Yes	Username or Email address.
password	string	Yes	Account password.

Request Payload (JSON):

```
{ "username": "tanvir@example.com", "password": "Password123!" }
```

Response Example (200/201 OK):

```
{ "user": { "id": 42, "username": "tanvir_japan", "full_name": "Tanvir Ahmed", "email": "tanvir@example.com", "is_staff": false, "profile": { "target_exam": "jft_basic", "japanese_level": "n4" } }, "tokens": { "access": "eyJhbGciOi...", "refresh": "eyJhbGciOi..." }, "message": "Welcome back, tanvir_japan!" }
```

POST /api/auth/google/

Auth: Public (None)

Description: Authenticates candidate via Google One-Tap or Google Sign-In (ID Token). Auto-registers if user does not exist.

Parameter	Type	Required	Description
id_token	string	Yes	Google ID Token JWT received from Google Identity Services SDK.

Request Payload (JSON):

```
{ "id_token": "eyJhbGciOiJSUzI1NiIsImtpZCI..." }
```

Response Example (200/201 OK):

```
{ "user": { "id": 45, "username": "tanvir_ahmed", "full_name": "Tanvir Ahmed", "email": "tanvir.google@gmail.com" }, "tokens": { "access": "...", "refresh": "...", "is_new_user": false, "message": "Welcome back, Tanvir!" }
```

POST	/api/auth/token/refresh/	Auth: Public (None)
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Description: Refreshes an expired JWT access token using the long-lived refresh token.

Parameter	Type	Required	Description
refresh	string	Yes	Valid Refresh Token string.

Request Payload (JSON):

```
{ "refresh": "eyJhbGciOi..." }
```

Response Example (200/201 OK):

```
{ "access": "eyJhbGciOi..." }
```

GET	/api/auth/me/	Auth: Required (Bearer)
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Description: Fetches authenticated candidate profile, full name, email, target exam, and Japanese level.

Response Example (200/201 OK):

```
{ "user": { "id": 42, "username": "tanvir_japan", "full_name": "Tanvir Ahmed", "email": "tanvir@example.com", "is_staff": false, "profile": { "bio": "Studying for JFT-Basic A2 exam in Dhaka.", "target_exam": "jft_basic", "target_exam_display": "JFT-Basic (A2 Standard)", "japanese_level": "n4", "japanese_level_display": "Elementary (N4 / A2)", "location": "Dhaka, Bangladesh" } } }
```

PUT / PATCH	/api/auth/profile/	Auth: Required (Bearer)
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Description: Updates candidate profile fields (bio, target exam, Japanese proficiency level, location).

Parameter	Type	Required	Description
bio	string	No	Short candidate bio / goals (max 500 chars).
target_exam	string	No	Choice: jft_basic, ssw_nursing, ssw_food, ssw_agri, ssw_const, etc.
japanese_level	string	No	Choice: beginner, n5, n4, n3, n2, n1.
location	string	No	City/Country (e.g. Dhaka, Bangladesh).

Request Payload (JSON):

```
{ "bio": "Targeting 220+ on Prometric JFT test next month.", "target_exam": "jft_basic", "japanese_level": "n4", "location": "Dhaka, Bangladesh" }
```

Response Example (200/201 OK):

```
{ "user": { ... }, "message": "Profile updated successfully!" }
```

GET	/api/auth/my-results/	Auth: Required (Bearer)
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Description: Returns candidate's full exam attempt history, pass rate %, highest scaled score (10–250), section stats, and graph chart series.

Response Example (200/201 OK):

```
{ "total_attempts": 5, "passed_attempts": 3, "pass_rate": 60, "highest_scaled_score": 218,
  "avg_scaled_score": 194, "highest_level": "A2.2 (A2)", "section_stats": [ { "key": "script_vocab", "name_en":
  "Script & Vocabulary", "correct": 48, "total": 60, "pct": 80 }, { "key": "conversation", "name_en":
  "Conversation & Expression", "correct": 45, "total": 60, "pct": 75 }, { "key": "listening", "name_en":
  "Listening Comprehension", "correct": 50, "total": 60, "pct": 83 }, { "key": "reading", "name_en": "Reading
  Comprehension", "correct": 42, "total": 60, "pct": 70 } ], "attempts": [ { "id": 108, "test_id": 1,
  "test_title": "JFT-Basic Official Practice Mock Test 1", "test_category": "basic", "score": 38,
  "total_questions": 42, "percentage": 90, "scaled_score": 226, "assessment_level": "A2.2 (A2)", "passed":
  true, "completed_at": "2026-08-24T18:30:00Z" } ], "chart_data": { "labels": ["Aug 20, 14:00", "Aug 22,
  10:30", "Aug 24, 18:30"], "scores": [180, 195, 226], "titles": ["Mock 1", "Mock 1", "Mock 1"] } }
```

2. Tests & CBT Examination Engine

Endpoints to list practice exams, retrieve structured CBT screens with native audio and 10-language translations, submit candidate answers, and view detailed scorecards.

GET	/api/tests/	Auth: Public (None)
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Description: Lists all published mock tests. Filterable by category (basic, skill). Staff/admin users also receive draft tests.

Parameter	Type	Required	Description
category	string	No	Filter tests: 'basic' (JFT-Basic) or 'skill' (SSW Skills).

Response Example (200/201 OK):

```
{ "tests": [ { "id": 1, "title": "JFT-Basic Official Practice Mock Test 1", "description": "Authentic
  60-minute Prometric simulation with 42 questions and audio listening.", "category": "basic",
  "requires_account": false, "is_published": true, "is_actual_exam_demo": true, "time_limit_seconds": 3600,
  "question_count": 42 }, { "id": 2, "title": "JFT-Basic Official Practice Mock Test 2", "category": "basic",
  "requires_account": true, "time_limit_seconds": 3600, "question_count": 42 } ], "tests_by_category": {
  "basic": [ ... ], "skill": [ ... ] } }
```

GET	/api/tests//	Auth: Public (None)
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Description: Retrieves detailed metadata for a single test (title, description, time limit, question count).

Response Example (200/201 OK):

```
{ "id": 1, "title": "JFT-Basic Official Practice Mock Test 1", "description": "Standard A2 evaluation test.",
  "category": "basic", "requires_account": false, "time_limit_seconds": 3600, "question_count": 42 }
```

GET

/api/tests//quiz/

Auth: Public (None)

Description: Retrieves structured CBT quiz screens/steps. If test.requires_account=true, Auth token is required. Questions include audio_url, image_url, and 10 language translation aids.

Parameter	Type	Required	Description
preview	string	No	Set '?preview=admin' to allow previewing unpublished draft tests.

Response Example (200/201 OK):

```
{ "test": { "id": 1, "title": "JFT-Basic Mock 1", "time_limit_seconds": 3600 }, "total_questions": 42, "total_steps": 35, "steps": [ { "step_number": 1, "section": "script_vocab", "group": null, "questions": [ { "id": 101, "type": "single_choice", "section": "script_vocab", "instruction": "1234", "resolved_instruction": "Choose the correct reading for the underlined word.", "prompt": "明日 雨が 降る。", "image_url": null, "audio_url": null, "translations": { "en": "Tomorrow it will rain.", "bn": "আমরা জানি কতটা", "vi": "Ngày mai trời sẽ mưa.", "id": "Besok akan turun hujan.", "my": "နောက်တော့ ရွာမှာ ရွာမှာ", "ne": "आइतको दिनमा आइतको", "th": "พรุ่งนี้จะมีฝนตก" }, "options": [ { "id": 401, "label": "A", "image_url": null, "order_index": 1 }, { "id": 402, "label": "B", "image_url": null, "order_index": 2 }, { "id": 403, "label": "C", "image_url": null, "order_index": 3 }, { "id": 404, "label": "D", "image_url": null, "order_index": 4 } ] } ] ] }
```

POST

/api/tests//submit/

Auth: Public (None)

Description: Submits candidate's quiz answers, calculates raw score, saves an Attempt record, and returns the attempt_id.

Parameter	Type	Required	Description
answers	object	Yes	Key-value map of question_id to selected answer_option_id.

Request Payload (JSON):

```
{ "answers": { "101": 401, "102": 406, "103": 412, "104": null } }
```

Response Example (200/201 OK):

```
{ "attempt_id": 158, "score": 35, "total_questions": 42, "message": "Quiz submitted successfully!" }
```

GET

/api/attempts//

Auth: Public (None)

Description: Returns official Prometric CEFR scorecard, 10–250 scaled score, pass/fail status, section breakdowns, and question-by-question review with correct answers.

Response Example (200/201 OK):

```
{ "attempt": { "id": 158, "score": 35, "total_questions": 42, "percentage": 83.3, "passed": true, "scaled_score": 210, "assessment_level": "A2.2 (A2)", "scaled_score_percent": 83.3, "completed_at": "2026-08-25T10:15:00Z" }, "test": { "id": 1, "title": "JFT-Basic Mock 1" }, "section_breakdown": { "script_vocab": { "name_ja": "単語", "name_en": "Script and Vocabulary", "correct": 10, "total": 12, "pct": 83 }, "conversation": { "name_ja": "会話", "name_en": "Conversation and Expression", "correct": 10, "total": 12, "pct": 83 }, "listening": { "name_ja": "聴解", "name_en": "Listening Comprehension", "correct": 8, "total": 10, "pct": 80 }, "reading": { "name_ja": "読解", "name_en": "Reading Comprehension", "correct": 7, "total": 8, "pct": 88 } }, "questions": [ { "id": 101, "prompt": "明日 雨が 降る。", "selected_option_id": 401, "is_answered_correctly": true, "options": [ { "id": 401, "label": "A", "is_correct": true }, { "id": 402, "label": "B", "is_correct": false } ] } ] }
```

3. Information & Venue Guides

Static and promotional data for JFT-Basic structure, Bangladesh Prometric test centers (BDJ01, BDJ02), and 12 SSW industry sectors.

GET	/api/info/jft/	Auth: Public (None)
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Description: Retrieves JFT-Basic overview, syllabus, Prometric Bangladesh test centers (BDJ01, BDJ02), address, map GPS coords, and voucher guide.

Response Example (200/201 OK):

```
{ "jft_info": { "title": "Japan Foundation Test for Basic Japanese (JFT-Basic)", "passing_score": "200 / 250 (Scaled Score)", "duration_minutes": 60, "questions_count": 42 }, "test_centers": [ { "code": "BDJ01", "name": "Prometric Testing Center - Dhaka Center 1", "address": "Banani, Dhaka, Bangladesh", "latitude": 23.7937, "longitude": 90.4066 }, { "code": "BDJ02", "name": "Prometric Testing Center - Dhaka Center 2", "address": "Dhanmondi, Dhaka, Bangladesh", "latitude": 23.7465, "longitude": 90.3760 } ] }
```

GET	/api/info/ssw/	Auth: Public (None)
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Description: Retrieves Specified Skilled Worker (SSW) visa requirements, 12 industry sectors (Nursing Care, Food Service, Agriculture, Construction, etc.), and venue guidelines.

Response Example (200/201 OK):

```
{ "ssw_info": { "title": "Specified Skilled Worker (SSW) Evaluation Exams" }, "ssw_sectors": [ { "id": "nursing", "name_en": "Nursing Care", "name_ja": "介護" }, { "id": "food_service", "name_en": "Food Service Industry", "name_ja": "食料品製造業" }, { "id": "agriculture", "name_en": "Agriculture", "name_ja": "農業" } ] }
```

4. Leaderboard & Candidate Profiles

Candidate rankings and public candidate showcase profiles with achievement badges.

GET	/api/leaderboard/	Auth: Public (None)
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Description: Returns candidate ranking list ordered by: 1) Tests Passed, 2) Highest Scaled Score, 3) Average Score, 4) Total Attempts. Splits into top 3 podium and rest.

Response Example (200/201 OK):

```
{ "total_candidates": 84, "top_three": [ { "rank": 1, "username": "fayaz_sensei", "full_name": "Azizul Hakim Fayaz", "passed_attempts": 12, "highest_score": 248, "avg_score": 238, "pass_rate": 100, "target_exam_display": "JFT-Basic" }, { "rank": 2, "username": "fayaz_sensei", "full_name": "Azizul Hakim Fayaz", "passed_attempts": 12, "highest_score": 248, "avg_score": 238, "pass_rate": 100, "target_exam_display": "JFT-Basic" }, { "rank": 3, "username": "fayaz_sensei", "full_name": "Azizul Hakim Fayaz", "passed_attempts": 12, "highest_score": 248, "avg_score": 238, "pass_rate": 100, "target_exam_display": "JFT-Basic" } ], "rankings": [ ... ], "current_user_rank": { "rank": 5, "highest_score": 218 } }
```

GET	/api/candidates//	Auth: Public (None)
-----	-------------------	---------------------

Description: Returns public profile for candidate by username, including unlocked badges (e.g. Registered, First Pass, High Scorer Elite, Podium Champion) and recent exam history.

Response Example (200/201 OK):

```
{ "id": 42, "username": "tanvir_japan", "full_name": "Tanvir Ahmed", "bio": "Preparing for JFT-Basic A2.",
"target_exam_display": "JFT-Basic", "japanese_level_display": "Elementary (N4 / A2)", "location": "Dhaka,
Bangladesh", "rank": 5, "stats": { "total_attempts": 5, "passed_attempts": 3, "pass_rate": 60,
"highest_scaled_score": 218, "avg_scaled_score": 194, "highest_level": "A2 (Passed)" }, "achievements": [ {
"id": "registered", "title": "Registered Candidate", "unlocked": true, "icon": "■" }, { "id": "first_pass",
"title": "JFT-Basic Qualified (A2)", "unlocked": true, "icon": "■" }, { "id": "high_scorer", "title": "High
Scorer Elite", "unlocked": false, "icon": "■" } ], "recent_attempts": [ ... ] }
```

5. Mobile App Integration Guide (Flutter / Dart Example)

Here is a production-ready HTTP Client and Authentication interceptor snippet in Dart / Flutter for mobile developers:

```
import 'dart:convert'; import 'package:http/http.dart' as http; import
'package:flutter_secure_storage/flutter_secure_storage.dart'; class ApiService { static const String baseUrl
= "https://gakkounoshiken.site/api"; static final _storage = FlutterSecureStorage(); // Helper to get stored
access token static Future<Map> _getHeaders({bool auth = false}) async { final headers = { 'Content-Type':
'application/json; charset=UTF-8', 'Accept': 'application/json', }; if (auth) { final token = await
_storage.read(key: 'jwt_access_token'); if (token != null) { headers['Authorization'] = 'Bearer $token'; } }
return headers; } // 1. Candidate Login static Future<Map> login(String usernameOrEmail, String password) async {
final response = await http.post( Uri.parse('$baseUrl/auth/login/'), headers: await _getHeaders(), body:
jsonEncode({'username': usernameOrEmail, 'password': password}), ); final data = jsonDecode(response.body);
if (response.statusCode == 200) { await _storage.write(key: 'jwt_access_token', value:
data['tokens']['access']); await _storage.write(key: 'jwt_refresh_token', value: data['tokens']['refresh']);
} return data; } // 2. Fetch CBT Quiz Steps static Future<Map> getQuizData(int testId) async { final response =
await http.get( Uri.parse('$baseUrl/tests/$testId/quiz/'), headers: await _getHeaders(auth: true), ); return
jsonDecode(utf8.decode(response.bodyBytes)); } // 3. Submit Answers static Future<Map> submitQuiz(int testId,
Map answers) async { final response = await http.post( Uri.parse('$baseUrl/tests/$testId/submit/'), headers:
await _getHeaders(auth: true), body: jsonEncode({'answers': answers}), ); return jsonDecode(response.body);
} }
```

6. HTTP Status Codes & Error Conventions

Code	Meaning	Description & Client Action
200 OK	Success	Request processed normally. Returns requested JSON payload.
201 Created	Resource Created	Returned upon successful registration, OTP verification, or quiz submission.
400 Bad Request	Validation Error	Missing parameters, password mismatch, or invalid OTP code. Inspect detail field.
401 Unauthorized	Auth Required	Missing or expired JWT access token. Attempt token refresh or redirect to login.
403 Forbidden	Permission Denied	Attempting to access draft tests or view other candidates' private results.
404 Not Found	Not Found	The specified test ID, attempt ID, or username does not exist.
500 Server Error	Internal Error	Unexpected backend exception. Client should prompt retry.